

Mergers and Counterfactual Prices

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Grad IO

Merger Simulation: Two Options

► Partial Merger Simulation

- Simulate a new price for p_j after acquiring good k holding the prices of all other goods (p_k, p_{-j}) fixed.
- Repeat for p_k and all other products involved in the merger.
- Compare price increases to **synergies** or cost savings.

► Full Merger Simulation

- Write down the full system of post-merger FOC.
- Adjust post-merger marginal costs for potential synergies.
- Solve for all prices at the new (post-merger) equilibrium (p_j, p_k, p_{-j}) .

Differentiated Products Bertrand

Recall the multi-product Bertrand FOCs:

$$\begin{aligned}\arg \max_{\mathbf{p} \in \mathcal{J}_f} \pi_f(\mathbf{p}) &= \sum_{j \in \mathcal{J}_f} (p_j - c_j) \cdot q_j(\mathbf{p}) \\ \rightarrow 0 &= q_j(\mathbf{p}) + \sum_{k \in \mathcal{J}_f} (p_k - c_k) \frac{\partial q_k}{\partial p_j}(\mathbf{p})\end{aligned}$$

It is helpful to define the matrix Δ with entries:

$$\Delta_{(j,k)}(\mathbf{p}) = \begin{cases} -\frac{\partial q_j}{\partial p_k}(\mathbf{p}) & \text{for } (j,k) \in \mathcal{J}_f \\ 0 & \text{for } (j,k) \notin \mathcal{J}_f \end{cases}$$

We can re-write the FOC in matrix form:

$$q(\mathbf{p}) = \Delta(\mathbf{p}) \cdot (\mathbf{p} - \mathbf{mc})$$

What does a merger do? **change the ownership matrix.**

- ▶ Step 1: Recover marginal costs $\widehat{\mathbf{mc}} = \mathbf{p} + \Delta(\mathbf{p})^{-1}q(\mathbf{p})$.
 - ▶ Step 1a: (Possibly) adjust marginal cost $\widehat{\mathbf{mc}} \cdot (1 - e)$ with some cost efficiency e .
 - ▶ Step 2: Change the ownership matrix $\Delta^{pre}(\mathbf{p}) \rightarrow \Delta^{post}(\mathbf{p})$.
 - ▶ Step 3: Solve for $\mathbf{p}^{post}$ via: $\mathbf{p} = \widehat{\mathbf{mc}} - \Delta(\mathbf{p})^{-1}q(\mathbf{p})$.
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- ▶ The first step is easy (just a matrix inverse).
 - ▶ The second step is trivial.
 - ▶ The third step is tricky because we have to solve an implicit system of equations. $\mathbf{p}$ is on both sides.

What does a merger do? **change the ownership matrix.**

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- ▶ Hold all other prices p_{-j} fixed at **pre-merger** prices.
- ▶ Adjust the marginal costs for potential efficiencies.
- ▶ Consider only the FOC for product j

$$0 = q_j(\mathbf{p}) + \sum_{k \in \mathcal{J}_f} (p_k - c_k) \frac{\partial q_k}{\partial p_j}(\mathbf{p})$$

- ▶ Solve for the new p_j given the change in the products controlled by firm f : $\mathcal{J}_f \rightarrow \mathcal{J}'_f$
- ▶ This is a single Gauss-Jacobi step (only products involved in merger).

Partial Merger Analysis: Why bother?

- ▶ We only need own and cross elasticities for products involved in the merger.
- ▶ Tends to show smaller price increases than full equilibrium merger analysis.
- ▶ Only solving a single equation rather than a system of J nonlinear equations.

How do we solve: $\mathbf{p} = \widehat{\mathbf{m}\mathbf{c}} - \Delta(\mathbf{p})^{-1}q(\mathbf{p})$?

1. Gauss Jacobi: Simultaneous Best Reply $p_j^{k+1}(\mathbf{p}_{-j}^k)$.
2. Gauss Seidel: Iterated Best Response $p_j^{k+1}(\mathbf{p}_{<j}^{k+1}, \mathbf{p}_{>j}^k)$.
3. Newton's Method: Set $\mathbf{p} - \widehat{\mathbf{m}\mathbf{c}} + \Delta(\mathbf{p})^{-1}q(\mathbf{p}) = 0$
but requires derivatives of $\Delta(\mathbf{p})^{-1}q(\mathbf{p})$
4. Fixed point iteration: $\mathbf{p} \leftarrow \widehat{\mathbf{m}\mathbf{c}} - \Delta(\mathbf{p})^{-1}q(\mathbf{p})$
 - Turns out this is **not a contraction**.
 - But you can get lucky... $\mathbf{p} - \widehat{\mathbf{m}\mathbf{c}} + \Delta(\mathbf{p})^{-1}q(\mathbf{p}) = 0$ means you have satisfied FOC's
5. Alternative fixed point.

Solving $F(\mathbf{p}) = 0$: The One-Slide Version

General problem: J nonlinear equations in J unknowns, $F(\mathbf{p}) = \mathbf{p} - \widehat{\mathbf{m}}\mathbf{c} + \Delta(\mathbf{p})^{-1}q(\mathbf{p}) = 0$.

Method	Update	Needs
Gauss-Jacobi	solve each $F_j(\mathbf{p}_j^{n+1}, \mathbf{p}_{-j}^n) = 0$	contraction + strong monotonicity
Gauss-Seidel	same, but use $\mathbf{p}_{<j}^{n+1}$ as you go	contraction + strong monotonicity
Newton-Raphson	$\mathbf{p}^{n+1} = \mathbf{p}^n - J_F(\mathbf{p}^n)^{-1}F(\mathbf{p}^n)$	Jacobian of $\Delta^{-1}q$; quadratic local convergence
Broyden / quasi-Newton	secant-style update of J_F^{-1}	no derivatives; slower
Picard	$\mathbf{p} \leftarrow \widehat{\mathbf{m}}\mathbf{c} - \Delta(\mathbf{p})^{-1}q(\mathbf{p})$	not a contraction (but you can get lucky)

- ▶ If Newton is feasible, start there. Best-response iteration is what most people reach for first and it is the least reliable.
- ▶ Derivations, convergence conditions, and accelerated fixed points (SQUAREM, Anderson) are in the **Systems of Equations** notes in Extra Notes.
- ▶ For the pricing game specifically there is a better option: exploit the structure of $\Delta(\mathbf{p})$ (next slide).

Exploit the (mixed) logit formula

For mixed logit with types i (weights w_i , price coefficients $\alpha_i > 0$), the Δ matrix in a single market is:

$$\Delta_{(j,k)}(\mathbf{p}) = -\frac{\partial \sigma_j}{\partial p_k} = \begin{cases} \sum_i w_i \alpha_i \sigma_{ij}(1 - \sigma_{ij}) & \text{if } j = k \\ -\sum_i w_i \alpha_i \sigma_{ij} \sigma_{ik} & \text{if } j \neq k \end{cases}$$

Both cases have the same structure, so Δ factors into a diagonal piece and a dense piece:

$$\Delta(\mathbf{p}) = \underbrace{\text{Diag}\left[\sum_i w_i \alpha_i \sigma_{ij}\right]}_{\Lambda(\mathbf{p})} - \underbrace{\left[\sum_i w_i \alpha_i \sigma_{ij} \sigma_{ik}\right]_{(j,k)}}_{\Gamma(\mathbf{p}) = \mathbf{S} \text{Diag}[w_i \alpha_i] \mathbf{S}'}$$

- $\mathbf{S}$ is $J \times I$ with columns $\boldsymbol{\sigma}_i$: Γ has **rank $\leq I$** (the same rank- I structure as the second-choice matrix in the micro-data week). Plain logit ($I = 1$): $\Lambda = \alpha \text{Diag}[\boldsymbol{\sigma}]$, $\Gamma = \alpha \boldsymbol{\sigma} \boldsymbol{\sigma}'$, rank one.
- Λ is diagonal, so the ownership matrix only masks Γ : $\mathcal{H} \odot \Delta = \Lambda - \mathcal{H} \odot \Gamma$.
- Price-specific coefficients (log price, income effects): replace α_i by $\alpha_{ik} = -\partial V_{ik} / \partial p_k$, indexed by the **column** k in Γ and by j in Λ . Then Γ is **no longer symmetric** (next slide).

Morrow Skerlos (2011) Fixed Point

Start from the FOC $(\mathcal{H} \odot \Delta)(\mathbf{p} - \mathbf{mc}) = \boldsymbol{\sigma}$, substitute the factorization, and rescale by Λ^{-1} :

$$(\mathbf{p} - \mathbf{mc}) \leftarrow \underbrace{\Lambda^{-1}(\mathbf{p}) \cdot (\mathcal{H} \odot \Gamma(\mathbf{p})) \cdot (\mathbf{p} - \mathbf{mc}) + \Lambda^{-1}(\mathbf{p}) \cdot \boldsymbol{\sigma}(\mathbf{p})}_{\zeta(\mathbf{p})}$$

- ▶ Reading: $\Lambda^{-1}\boldsymbol{\sigma}$ has entries $\sigma_j / \sum_i w_i \alpha_i \sigma_{ij}$, the markup ignoring cannibalization; the Γ term adds back own cannibalization (Γ_{jj}) and the opportunity cost of sister products. Single-product plain logit check:
 $(p_j - mc_j)(1 - \sigma_j) = 1/\alpha$.
- ▶ Each iteration needs only σ_{ij} at current prices: no derivatives, no Jacobian inverse, Λ^{-1} is a diagonal scaling. Converges fast and stably, far better than Gauss-Seidel or Gauss-Jacobi on the raw FOC. Honestly, the best way to solve large pricing games; PyBLP's default.
- ▶ Orientation: the FOC for p_j needs $\partial\sigma_k/\partial p_j$, so in general the mask multiplies the **transpose**, $\mathcal{H} \odot \Gamma'$. Irrelevant with common α_i (symmetric Γ) and symmetric $\mathcal{H}$; it matters with product-specific α_{ik} or asymmetric κ_{fg} (conduct lecture).
- ▶ With $\alpha_i < 0$ as the signed coefficient the sign on $\Lambda^{-1}\boldsymbol{\sigma}$ flips; that is how Conlon Gortmaker (2020) write it.

Lots of cases where we want to recompute prices:

- ▶ Change conduct/ownership $\mathcal{H}_t(\kappa)$.
- ▶ Change the mc (add a tax, tariff, etc.)
- ▶ Add/drop a product/products from the choice set $\mathcal{J}_t$.
- ▶ Change demographics of consumers d_i